

TIMOTHY MICHELETTO

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A seasoned ML Engineer with a rare combination of two decades of software engineering experience and 4+ years of specialised ML/AI experience at the Staff Engineer and Engineering Manager level. Standout strength in ML Ops, specifically deploying and operationalising ML systems in production environments. Looking for an opportunity to accelerate AI product development with strong ML platform engineering experience.

SKILLS

Languages: Python, SQL, HCL

Tools & Frameworks: AWS, Prefect, Snowflake, Kubernetes, Terraform, Git, Docker, Dask, FastAPI, Pandas, Postgres

Applications: ML Ops, Machine Learning, LLMs, GenAI

PROFESSIONAL EXPERIENCE

XERO

Melbourne, Australia

Senior Engineer - Machine Learning

September 2024 - September 2025

- Engineered an aggregation model to select the best automated bank reconciliation predictions from an ensemble of specialist models, according to business rules implemented in Python.
- Optimised a model evaluation pipeline to reliably process several terrabytes of data on a single EC2 instance utilising Dask distributed and parrallel batch processing which reduced the processing time from > 6 hours to < 2, significantly shortening the feedback loop for data scientists.
- Refactored a Levenstein distance based algorithm implemented in Python and Pandas, using Cython and other high performance Python programming techniques to improve performance by 350%.
- Implemented ETL pipelines in SQL, Python and Pandas to produce training and evaluation datasets in Snowflake, enabling automated deterministic training and evaluation of models.

XERO

Melbourne, Australia

Engineering Manager - Machine Learning

September 2023 - September 2024

- Led the end-to-end engineering delivery of a RAG based customer support agent, Xero's first implementation of GenAI, coordinating cross-functional alignment between Applied Science, Engineering and Customer Experience teams.
- Designed and executed the organizational restructuring of a 14-engineer team into 3 domain-aligned pods, resulting in higher levels of motivation, happiness and productivity in the newly formed pods.
- Directly managed and mentored a multi-disciplinary team of 9 engineers; championed professional development and refined engineering hiring practices to increase talent density and team cohesion.
- Championed a number of AI platform initiative, standardising architectural patterns to reduce fragmentation, technical debt and increase delivery velocity.
- Directed the technical implementation of Right to be Forgotten' workflows across all AI products data stores, ensuring GDPR compliance goals for the Data and AI org were met with zero production incidents.

XERO

Melbourne, Australia

Staff Engineer - Data Applications

March 2022 - September 2023

- Architected and led the implementation of the Cash flow Understanding and Prediction Service (CUPS) from data ingestion to prediction serving, utilising Snowflake, Prefect, Kubernetes, AWS DynamoDB and FastAPI.
- Led the implementation distributed batch inference pipelines processing terabytes of data daily, generating cash flow predictions for all of Xero's > 2M customers, utilising Prefect, Dask-Kubernetes, AWS DynamoDB
- Led the Implementation of low latency serving and observability including SLOs/SLIs of predictions utilising FastAPI, AWS DynamoDB, AWS Kinesis, Snowflake
- Developed reusable Terraform modules to provision ML inference and serving infrastructure on AWS including IAM roles, S3 Buckets, DynamoDB tables, Secrets and Kinesis streams
- Championed cross-functional collaboration between Product and Applied Science, serving as the primary technical lead to ensure research-driven initiatives met engineering scalability and product-market goals.

XERO

Melbourne, Australia

Senior Engineer - Data Applications

May 2021 - March 2022

- Accelerated the deployment of the cash flow forecasting model by building automated CI/CD pipelines for Python packages, reducing deployment friction between Science and Engineering.
- Performed model enhancements and evaluation on cash flow forecasting model.

LATITUDE FINANCIAL SERVICES

Engineering Lead

Melbourne, Australia

April 2020 - May 2021

- Spearheaded the architectural transformation of the Identity Platform utilising Okta, Golang, enabling Single Sign-On (SSO) for millions of customers.
- Collaborated with Engineering Leads across the business to identify cross cutting concerns to incorporate into our platform capability.
- Executed the migration of legacy user bases from AWS Cognito to Okta with zero downtime.
- Acted as Technical Product Manager for a platform team of 4 engineers, defining roadmap requirements and driving execution.

SEEK

Principal Software Developer

Melbourne, Australia

July 2018 - March 2020

- Replaced decades old, deeply embedded legacy middleware with simple AWS Lambda functions utilising C#.
- Core contributor to new Job posting API replacing the existing monolith that facilitates publishing Jobs on the SEEK website, utilising TypeScript, NodeJS and AWS DynamoDB.
- Led the implementation of a Resume Parsing pipeline utilising Scala, AWS Lambda and AWS Kinesis to parse information from resumes uploaded by candidates applying for jobs and update their SEEK profile.
- Led the implementation of a recommendation system to estimate and confirm candidate's expected salary and their working rights utilising Scala, Airflow, SQL and AWS Lambda.

SEEK

Senior Software Developer

Melbourne, Australia

March 2016 - July 2018

- Developed event-driven microservices utilising NodeJS, AWS SNS/SQS to power real-time candidate feedback loops.
- Integrated complex variable pricing logic into high-traffic C# ASP.NET monoliths.

LEICA BIOSYSTEMS

Senior Software Engineer

Melbourne, Australia

May 2013 - March 2016

- Developed IoT-style monitoring dashboards for pathology laboratory equipment using AngularJS and C#.
- Built the Inventory Manager application to optimize consumable reordering for lab supervisors.

EARLY CAREER

Various Roles: Integration Consultant, Lead Developer, Graduate

Australia & UK

2003 - 2012

- Enterprise Integration Consultant | Various Contracts (UK) (2008 - 2012)
- Lead Integration Developer | Linfox (2007 - 2008)
- Consultant / Graduate | Oakton / Downer (2003 - 2006)

EDUCATION

RMIT University

Bachelor of Engineering (Computer Systems)

Melbourne, Australia

2000 - 2002